Chapter 3 Review

Campbell Biology, AP Edition, 12e

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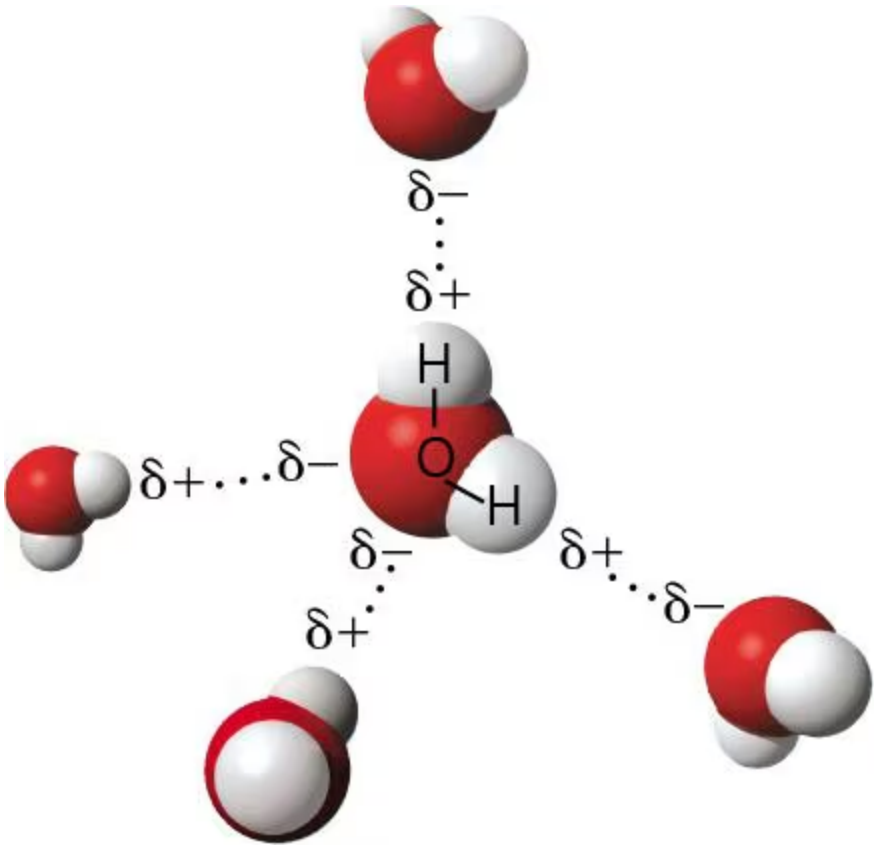
Summary of Key Concepts

How do the polar covalent bonds that form between oxygen and hydrogen atoms in a water molecule contribute to all the different emergent properties of water upon which living systems depend? **(BIG IDEA 2)**

Concept 3.1: Polar covalent bonds in water molecules result in hydrogen bonding

- Water is a **polar molecule**. A hydrogen bond forms when a partially negatively charged region on the oxygen of one water molecule is attracted to the partially positively charged hydrogen of a nearby water molecule. Hydrogen bonding between water molecules is the basis for water's properties.



Hydrogen bonds between water molecules. The charge regions in a water molecule are due to its polar covalent bonds. Oppositely charged regions of neighboring water molecules are attracted to each other, forming hydrogen bonds. Each molecule can hydrogen-bond to a maximum of four other water molecules, and these associations are constantly changing.

DRAW IT*Label a hydrogen bond and a polar covalent bond in the diagram of five water molecules. Is a hydrogen bond a covalent bond? Explain.*

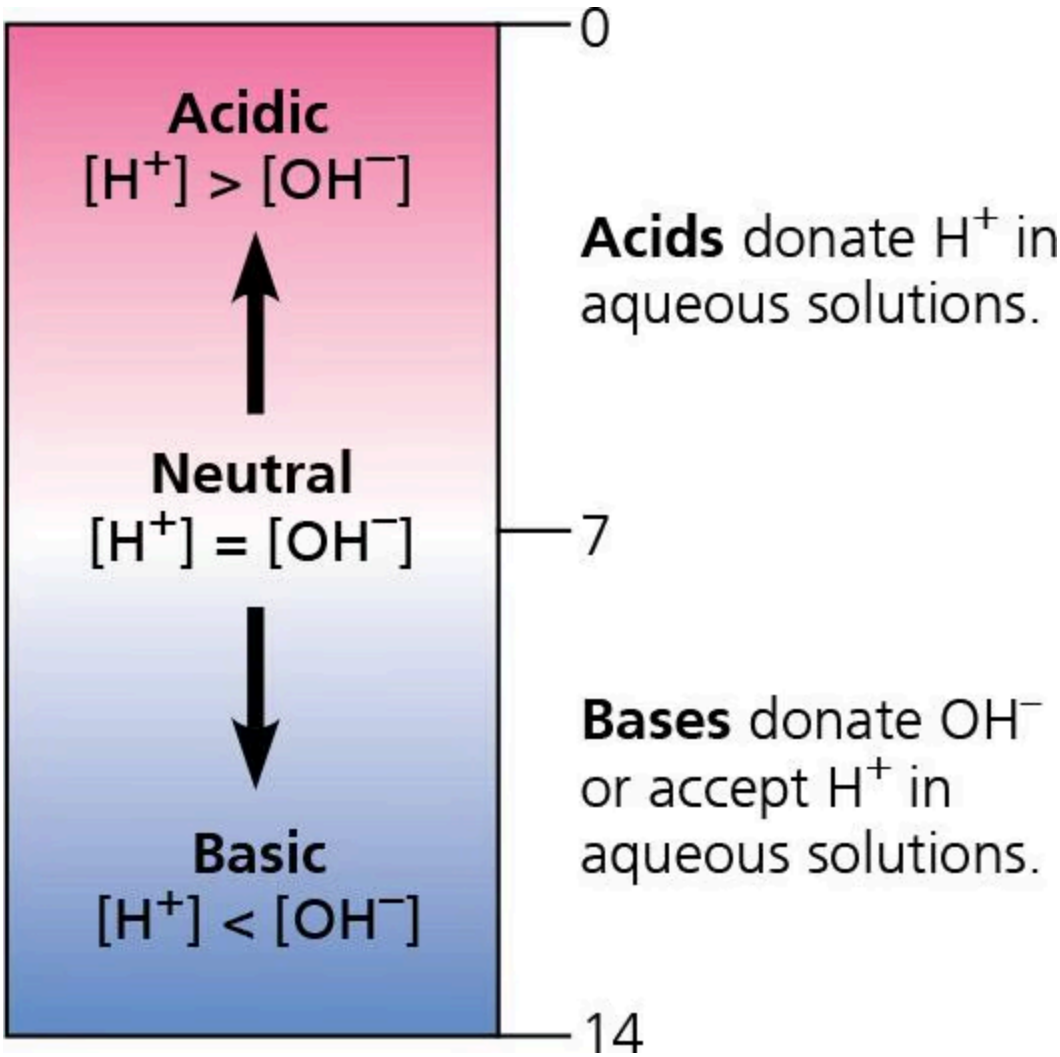
Concept 3.2: Four emergent properties of water contribute to Earth's suitability for life

- Hydrogen bonding keeps water molecules close to each other, giving water **cohesion**. Hydrogen bonding is also responsible for water's **surface tension**.
- Water has a high **specific heat**: Heat is absorbed when hydrogen bonds break and is released when hydrogen bonds form. This helps keep **temperatures** ****relatively steady, within limits that permit life. **Evaporative cooling** ****is based on water's high **heat of vaporization**. The evaporative loss of the most energetic water molecules cools a surface.
- Ice floats because it is less dense than liquid water. This property allows life to exist under the frozen surfaces of lakes and polar seas.
- Water is an unusually versatile **solvent**because its polar molecules are attracted to ions and polar substances that can form hydrogen bonds. **Hydrophilic** substances have an affinity for water; **hydrophobic**substances do not. **Molarity**, the number of moles of **solute** per liter of **solution**, is used as a measure of solute concentration in solutions. A **mole** is a certain number of molecules of a substance. The mass of a mole of a substance in grams is the same as the **molecular mass** in daltons.
- The emergent properties of water support life on Earth and may contribute to the potential for life to have evolved on other planets.

Describe how different types of solutes dissolve in water. Explain what a solution is.

Concept 3.3: Acidic and basic conditions affect living organisms

- A water molecule can transfer an H+ to another water molecule to form H3O+ (represented simply by H+) and OH−.
- The concentration of H+ is expressed as ****pH; pH=−log [H+]. A **buffer** consists of an acid-base pair that combines reversibly with hydrogen ions, allowing it to resist pH changes.



p H Scale from 0 to 14. 7 to 0 is Acidic (H plus is greater than O H minus), 7 is neutral (H plus = O H minus), 7 to 14 is basic (H plus is less than O H minus). Acids donate H plus in aqueous solutions. Bases donate O H minus or accept H plus in aqueous solutions.

- The burning of fossil fuels increases the amount of CO2 in the atmosphere. Some CO2 dissolves in the oceans, causing **ocean acidification**, which has potentially grave consequences for marine organisms that rely on calcification.

*Explain what happens to the concentration of hydrogen ions in an aqueous solution when you add a base and cause the concentration of*OH−*to rise to*10−3*. What is the*pH*of this solution?*